Eng. Haitham Abdullah Hussein

PRODUCTION & MECHANICAL DESIGN ENGINEERING

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PROFESSIONAL SUMMARY

Ambitious Mechanical Engineering student at Kafrelsheikh University specialized in the Production and Mechanical Design Department (Credit Hour System). Possess a robust technical foundation and a strong passion for hands-on applications in Mechanical Design, Engineering Simulation, and Computer-Aided Engineering (CAD/CAE). Extensively trained in Mechanical, Electrical, and Plumbing (MEP) systems with a dedicated focus on HVAC design and firefighting simulations. Proficient in integrating AI platforms to accelerate scientific research and engineer highly precise, optimized design solutions.

EDUCATION

Bachelor of Science in Mechanical Engineering SEP 2022 — PRESENT

Kafrelsheikh University — Faculty of Engineering

- Specialization: Production and Mechanical Design Department (Credit Hour System).
- Core Focus: Thermodynamics, Fluid Mechanics, Machine Component Design, and FEA/Structural Analysis.

TECHNICAL CERTIFICATIONS

Dassault Systèmes:
- Certified SOLIDWORKS Professional (CSWP) – Mechanical Design
- xDesign Associate & xShape Associate
- xMold Associate
- Sustainability Associate
- Additive Manufacturing Associate
- 3DEXPERIENCE 3DSwymer Associate

Ansys:
- Ansys Fluent Meshing Watertight Geometry Workflow

ENGINEERING PROJECTS

CFD Simulation of Vertical Axis Wind Turbines (VAWT) — MARCH 2026

- Executed advanced Computational Fluid Dynamics (CFD) simulations for a VAWT utilizing **NACA 0018 airfoils**.
- Generated and optimized the computational mesh, effectively controlling grid quality parameters including a Skewness factor of 0.54 and strictly resolving Y+ criteria to ensure hydrodynamic accuracy.

Static Structural Analysis of a Formula Student Chassis — MARCH 2026

- Performed static structural finite element analysis (FEA) on a Formula Student race car chassis engineered from **Chromoly 4130 steel** using ANSYS Mechanical.
- Conducted torsional rigidity benchmarking, established precise boundary conditions, and resolved structural stress-concentration zones.

Feasibility Study & Engineering Design for an MEP & HVAC Services Office — MARCH 2026

- Developed a comprehensive technical, financial, and market feasibility framework for establishing an engineering consultancy firm.
- Incorporated international HVAC design codes, focusing on system classifications for supply/return air and duct/pipe sizing calculations.

TECHNICAL SKILLS

CAD/CAE Tools:	SOLIDWORKS, ANSYS Fluent, Ansys CFX, Ansys Mechanical, Fluid Dynamics & Mesh Optimization (Y+, Skewness).
MEP & HVAC Design:	Revit MEP (Family Creation, System Classifications), Duct Sizing, Pipe Sizing, Ventilation Design.
AI Integration & Research:	Bohrium AI Mentor, Google NotebookLM, mnml.ai, Google AI Pro.

FIELD EXPERIENCE

Technical HVAC Engineering Site Visit — MARCH 2, 2026

Ain Shams University Project Site

- Participated in an immersive site visit analyzing large-scale central HVAC infrastructure, air handling units (AHUs), ductwork routing, and real-world construction installation alignments.